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Effect of PBX Prill Size on the Compressive Properties of PBX 9502

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Abstract. Plastic-Bonded Explosive (PBX) prills are agglomerates a few millimeters in diameter composed of High Explosive crystals non-uniformly distributed in and surrounded by a polymer binder. When pressed at elevated temperatures at or above the binder melt temperature the polymer flows to uniformly coat the HE crystals to create a well-consolidated compaction. In reality it does so imperfectly, such that x-ray tomographic scans often look like a collection of prills mashed together. It is speculated that the larger the prill, the more non-uniform the binder distribution, but this has never been proven. Even if that were not so, the larger the prill the farther binder must flow toward the middle to homogenize during pressing. Thus, the degree of homogenization depends in part on prill size. The degree to which binder flows during pressing in turn affects the void distribution within pressed charges, which one suspects will affect shock sensitivity and material strength. In this paper we explore, for ~30 PBX 9502 formulation batches using the same TATB powder lot, how prill size affects the compressive mechanical properties.

INTRODUCTION

PBX (plastic-bonded explosive) 9502 is comprised of 95 wt% TATB (1,3,5-triamino-2,4,6-trinitrobenzene) explosive crystals and 5 wt% Kel-F [1], a fluoro co-polymer, as the binder. This PBX material is formulated using a wet slurry process which produces agglomerates of TATB crystals surrounded by Kel-F binder known as prills. These prills vary in size depending on a variety of processing variables. One can imagine that as the prill size increases there is less binder toward the middle of the TATB agglomeration due to the binder not flowing all the way to the center during the slurry process. This potentially leads to prills with a low binder, solid rich center instead of evenly coated crystals throughout the prill.

To press the material, prills are heated to within or just above the melt range of the polymer so it flows to better meld the prills together into a single consolidated part. For Kel-F the melt temperature ranges from 80 to 100°C depending on the level of crystallinity [2]. Although the binder flows, one can imagine that if the prills have low binder content toward the center of larger prill sizes, the binder may not evenly distribute toward the center of prills during pressing. This could result in a consolidated part that is not well melded together, affecting the overall strength and stiffness of the material.

It is known that particle size distribution also affects the ability of a material to compact together well [3-4]. Specifically, optimized particle size distributions will have a spread of sizes where smaller particles will fill in void space between larger particles which increases density of the powder and consolidated material. Density of a consolidated part is known to affect the mechanical properties, usually resulting in increased overall strength and stiffness of the material. One can also imagine that the presence of smaller particles between larger ones during the pressing process of a PBX will help meld the prills together better than prills of just one size. Using our knowledge of prill formation, binder flow during pressing, and compaction theory, we investigate how prill size affects the compressive parameters of 29 batches of PBX 9502.

EXPERIMENTAL PROCEDURE

Particle Size Distribution Measurements

The particle size distribution (PSD) measurements for all 29 batches of PBX 9502 were performed on a Camsizer X-2 using the X-Fall Module with cascade method. A small sample of each batch was run through the Camsizer three times and the data averaged. Figure 1 shows the three-run average volume distribution ($q_3[x]$) for each batch versus the particle size (x). PBX 9502 prill size for all 29 batches generally peaks below ~ 2 mm with a few lots larger than 2 mm. Most of the lots have strong peaks at one particular size meaning most of the prills in that batch are of that size. A few of the batches have much broader peaks and tend to be on the larger side compared to the other batches.

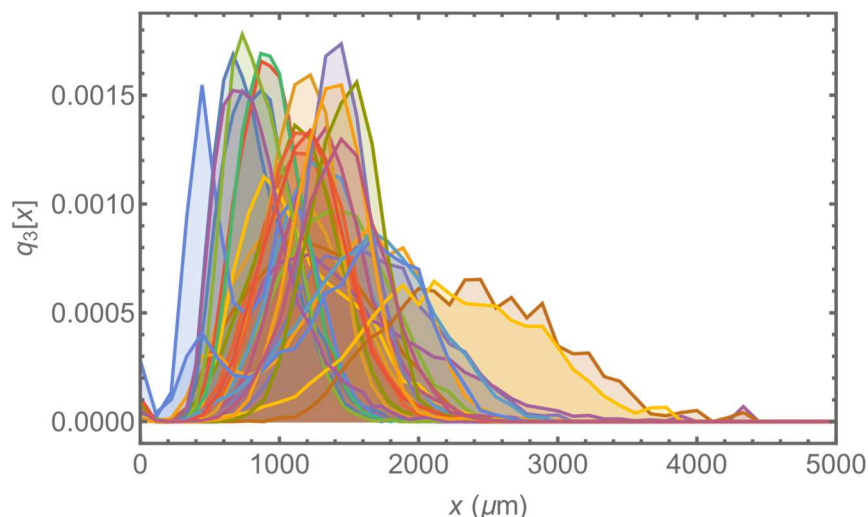


FIGURE 1. Average volume distribution ($q_3[x]$) versus prill size (x) of all 29 batches of PBX 9502.

Pressing Specimens

Each batch of PBX 9502 was die-pressed into compression cylinders. Material was heated to 90°C and pressed to a density range of $1.89 \pm 0.005 \text{ g/cm}^3$. All cylinders were 0.75 inches in length and 0.5 inches in diameter. Density of the specimens was measured using immersion methods and parts desiccated for 2 full weeks before compression testing.

Compression Testing

Compression tests were performed on an Instron 5567 retrofitted with an MTS ReNew/E controller and test temperature controlled with a Bemco, Inc. environmental chamber to within $\pm 0.2^{\circ}\text{C}$. Strain on the specimen was measured using two oppositely mounted knife-edge extensometers, Instron model 826-2026. All compression tests were performed at 50°C and 0.075 in/min to achieve a near constant strain rate of 0.001667 in/in/sec.

RESULTS AND DISCUSSION

From the raw stress vs. strain curves (not shown here), three parameters can be obtained to describe the compressive mechanical response. Ultimate Stress (σ_m), in MPa, is the maximum stress achieved at failure, Strain at Ultimate Stress (ϵ_m), in %, is the corresponding strain value at Ultimate Stress, and Modulus (E), in MPa/%, is the slope of the line tangent to the rising edge of the stress-strain curve. Ideally, the modulus represents the linear and elastic portion the curve in the low strain region. However, explosive composites tend to have complex behavior with no linear region. For this work, the modulus is defined to be the linear fit to a similar portion of the stress-strain curve

centered at 25% of the Ultimate Stress (E_{25}). The accuracy of the three parameter values is directly related to the systematic errors and nature of the quasi-static measurements. The ultimate stress and strain at ultimate stress are obtained with the highest accuracy. The value of the elastic modulus is highly dependent on how the term is defined (i.e., where on the stress-strain curve the slope is calculated.). By following an established protocol (i.e., 25% of ultimate stress) the parameter is well-defined and does not show significant statistical variation under a given set of experimental conditions. Figure 2 shows the three compressive parameters plotted against the PSD for each batch. The Arithmetic Mean for each curve shown in Figure 1 is used to represent the PSD as a single value for each batch. A linear fit to each parameter is shown in Figure 2 along with the 95% confidence interval.

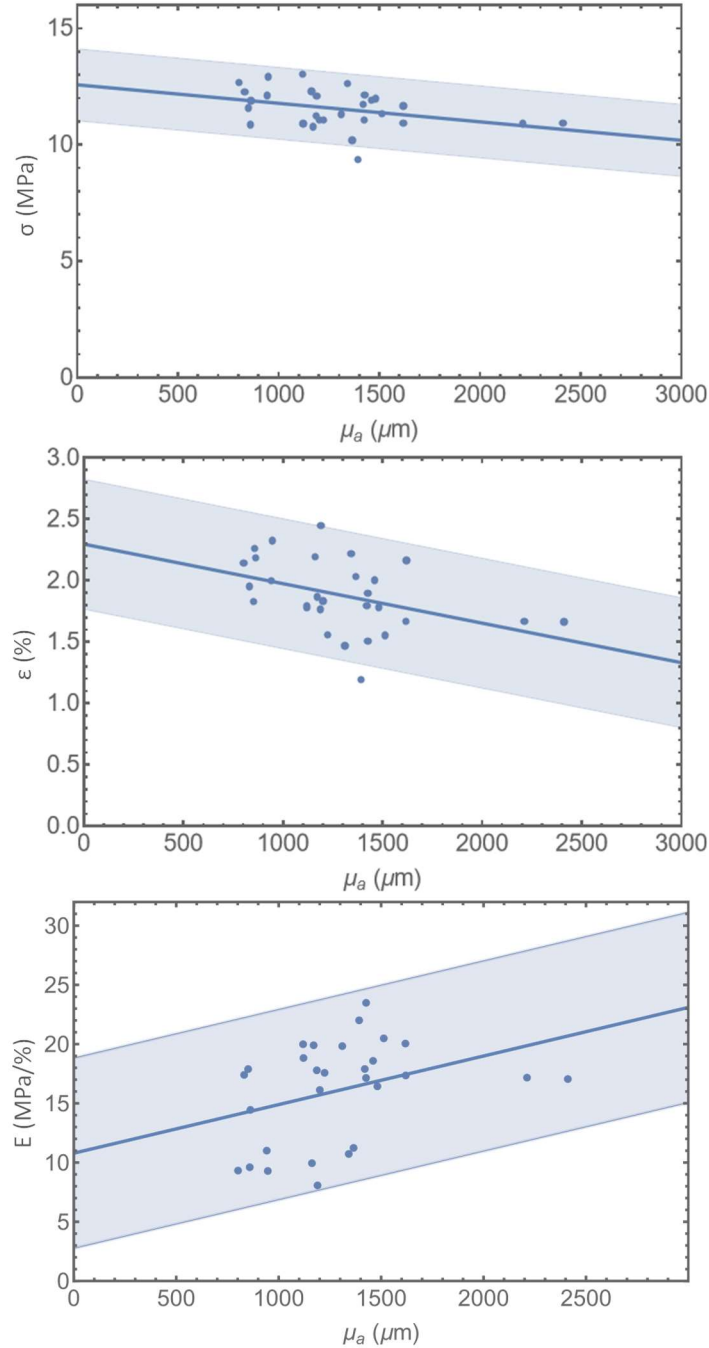


FIGURE 2. Ultimate Stress (top), Strain at Ultimate Stress (middle), and Modulus at 25% of Ultimate Stress (bottom) versus the Particle Size for 29 batches of PBX 9502.

The general trends in Figure 2 show that σ_m decreases, ϵ_m decreases, and E_{25} increases as prill size increases. Generally, this means the material exhibits stiffer behavior but is weaker as prill size increases. However, these relationships to prill size are not very strong, exhibited by the statistics reported in Table 1 for each parameter. The R-value, or the correlation coefficient, for all three parameters is less than 0.5 (sign independent) meaning there is very little correlation between prill size and each parameter. The R^2 -value, or coefficient of determination, is very low for all three parameters, between 1 and 2%, meaning the variation in the values for each of the three parameters is not related to the variation in the prill size. This indicates that each of the compressive parameters are probably not dependent on prill size alone. Other processing factors of each individual batch should be considered and evaluated as potential causes for the variation in particle size and resulting compressive properties.

TABLE 1. R-value, R^2 -value, F-Value, and p-value for the trends shown in Figure 2 for all three compressive parameters.

Parameter	R-Value	R^2 -Value	F-Value	p-Value
Ultimate Stress	-0.36	0.127	3.94	0.057
Strain @ Ultimate Stress	-0.41	0.170	5.54	0.026
Modulus @ 25% of Ultimate Stress	0.36	0.126	3.90	0.059

CONCLUSIONS

The relationship between PBX 9502 prill size and mechanical properties were investigated via compression testing and PSD measurements. There is a general weakening and stiffening effect as the Arithmetic means of the PSD increases. However, the three compressive parameters do not exhibit strong correlate with PSD. Other batch dependent factors should be investigated before further conclusions can be made.

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